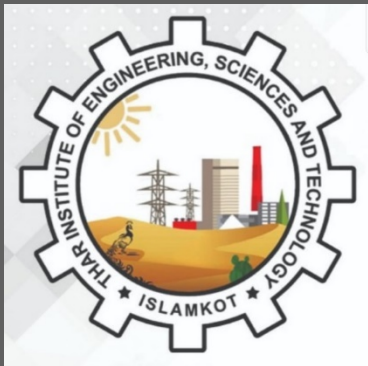




Thar Institute Of Engineering Science and Technology

NED University of Engineering and Technology



MT-331 PROBABILITY & STATISTICS

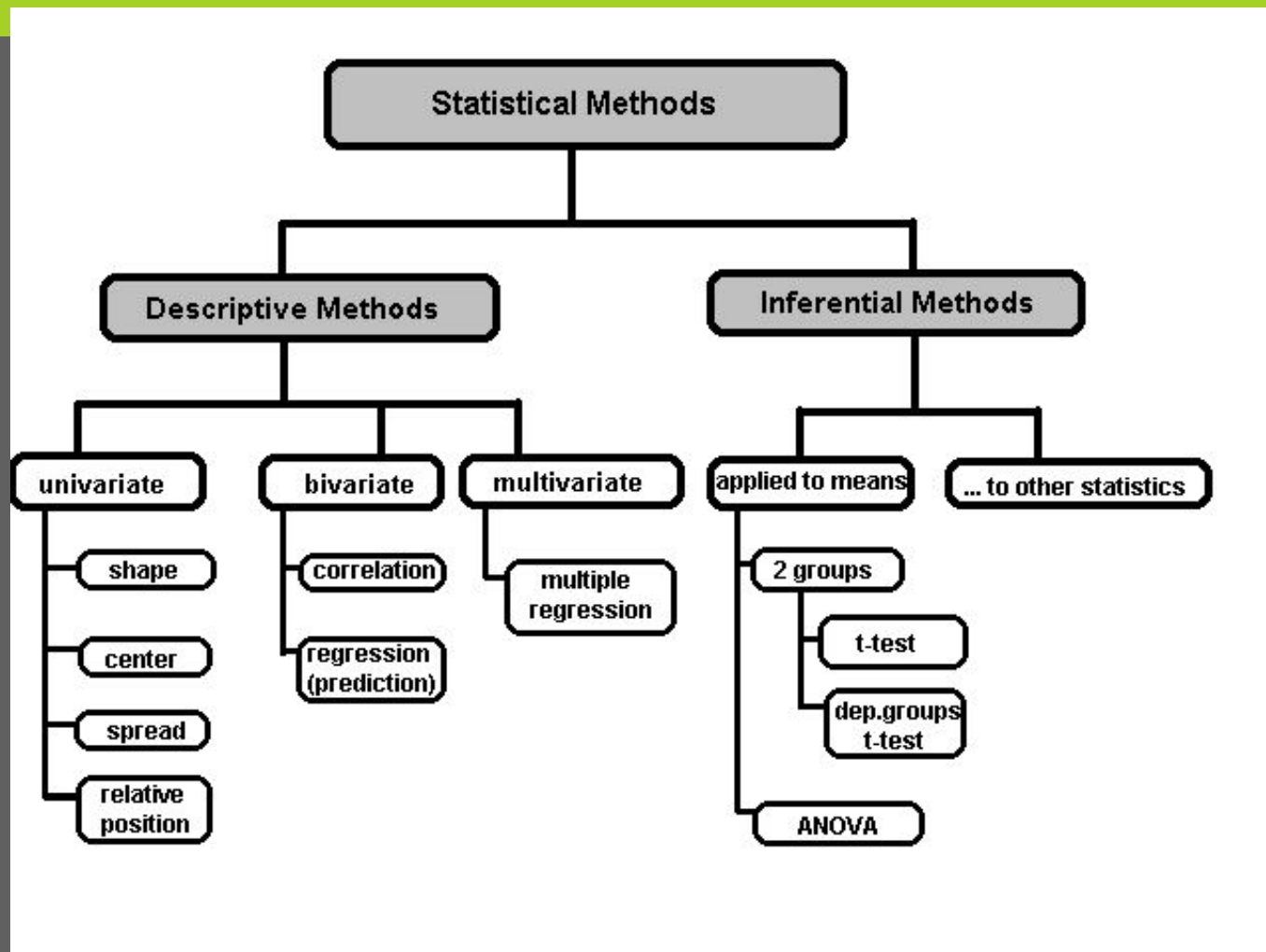
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MS- Applied Mathematics

BASICS OF STATISTICS

Definition: Science of collection, presentation, analysis, and reasonable interpretation of data.

Statistics presents a rigorous scientific method for gaining insight into data. For example, suppose we measure the weight of 100 patients in a study. With so many measurements, simply looking at the data fails to provide an informative account. However statistics can give an instant overall picture of data based on graphical presentation or numerical summarization irrespective to the number of data points. Besides data summarization, another important task of statistics is to make inference and predict relations of variables.

A TAXONOMY OF STATISTICS



STATISTICAL DESCRIPTION OF DATA

- Statistics describes a numeric set of data by its
 - Center
 - Variability
 - Shape
- Statistics describes a categorical set of data by
 - Frequency, percentage or proportion of each category

SOME DEFINITIONS

Variable - any characteristic of an individual or entity. A variable can take different values for different individuals. Variables can be *categorical* or *quantitative*. Per S. S. Stevens...

- **Nominal** - Categorical variables with no inherent order or ranking sequence such as names or classes (e.g., gender). Value may be a numerical, but without numerical value (e.g., I, II, III). The only operation that can be applied to Nominal variables is enumeration.
- **Ordinal** - Variables with an inherent rank or order, e.g. mild, moderate, severe. Can be compared for equality, or greater or less, but not *how much* greater or less.
- **Interval** - Values of the variable are ordered as in Ordinal, and additionally, differences between values are meaningful, however, the scale is not absolutely anchored. Calendar dates and temperatures on the Fahrenheit scale are examples. Addition and subtraction, but not multiplication and division are meaningful operations.
- **Ratio** - Variables with all properties of Interval plus an absolute, non-arbitrary zero point, e.g. age, weight, temperature (Kelvin). Addition, subtraction, multiplication, and division are all meaningful operations.

SOME DEFINITIONS

Distribution - (of a variable) tells us what values the variable takes and how often it takes these values.

- Unimodal - having a single peak
- Bimodal - having two distinct peaks
- Symmetric - left and right half are mirror images.

FREQUENCY DISTRIBUTION

Consider a data set of 26 children of ages 1-6 years. Then the frequency distribution of variable 'age' can be tabulated as follows:

Frequency Distribution of Age

Age	1	2	3	4	5	6
Frequency	5	3	7	5	4	2

Grouped Frequency Distribution of Age:

Age Group	1-2	3-4	5-6
Frequency	8	12	6

CUMULATIVE FREQUENCY

Cumulative frequency of data in previous page

Age	1	2	3	4	5	6
Frequency	5	3	7	5	4	2
Cumulative Frequency	5	8	15	20	24	26

Age Group	1-2	3-4	5-6
Frequency	8	12	6
Cumulative Frequency	8	20	26

DATA PRESENTATION

Two types of statistical presentation of data - graphical and numerical.

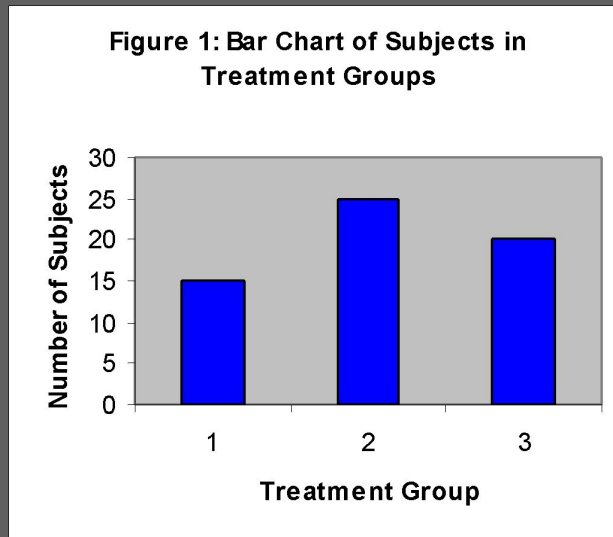
Graphical Presentation: We look for the overall pattern and for striking deviations from that pattern. Over all pattern usually described by shape, center, and spread of the data. An individual value that falls outside the overall pattern is called an ***outlier***.

Bar diagram and Pie charts are used for categorical variables.

Histogram, stem and leaf and Box-plot are used for numerical variable.

Data Presentation –Categorical Variable

Bar Diagram: Lists the categories and presents the percent or count of individuals who fall in each category.

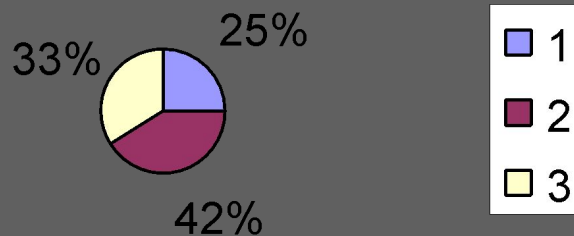


Treatment Group	Frequency	Proportion	Percent (%)
1	15	$(15/60)=0.25$	25.0
2	25	$(25/60)=0.333$	41.7
3	20	$(20/60)=0.417$	33.3
Total	60	1.00	100

Data Presentation –Categorical Variable

Pie Chart: Lists the categories and presents the percent or count of individuals who fall in each category.

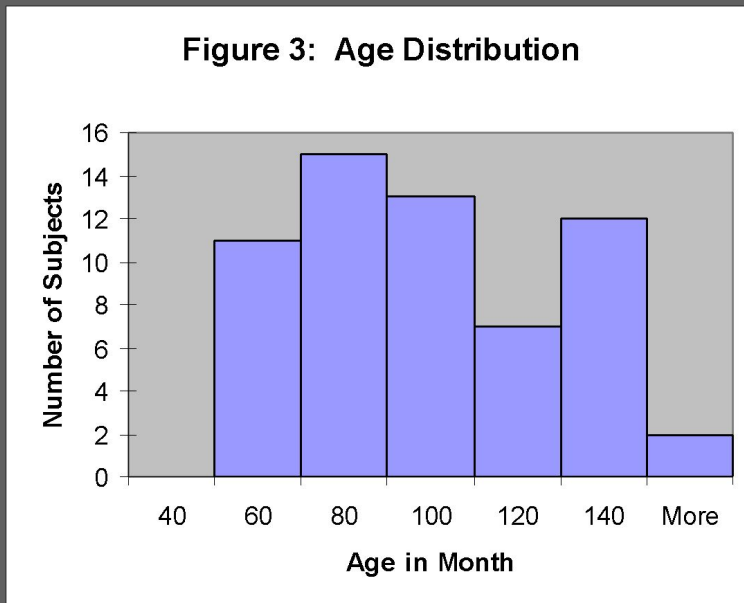
Figure 2: Pie Chart of Subjects in Treatment Groups



Treatment Group	Frequency	Proportion	Percent (%)
1	15	$(15/60)=0.25$	25.0
2	25	$(25/60)=0.333$	41.7
3	20	$(20/60)=0.417$	33.3
Total	60	1.00	100

GRAPHICAL PRESENTATION –NUMERICAL VARIABLE

Histogram: Overall pattern can be described by its **shape**, **center**, and **spread**. The following age distribution is **right skewed**. The **center** lies between **80 to 100**. **No outliers**.

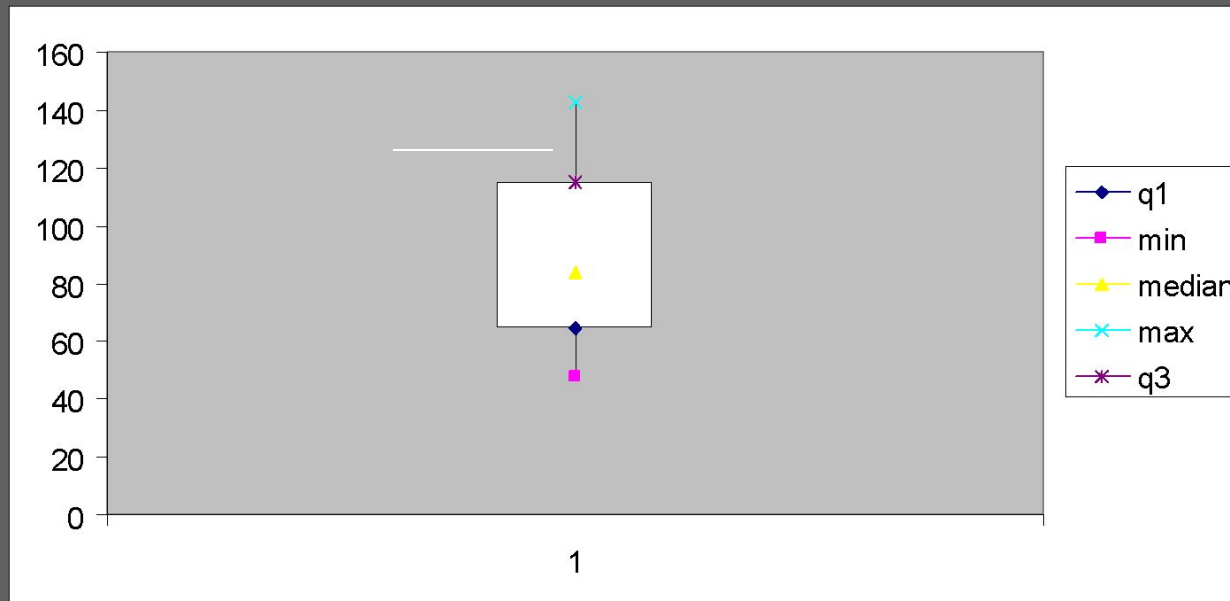


Mean	90.41666667
Standard Error	3.902649518
Median	84
Mode	84
Standard Deviation	30.22979318
Sample Variance	913.8403955
Kurtosis	-1.183899591
Skewness	0.389872725
Range	95
Minimum	48
Maximum	143
Sum	5425
Count	60

GRAPHICAL PRESENTATION –NUMERICAL VARIABLE

Box-Plot: Describes the five-number summary

Figure 3: Distribution of Age

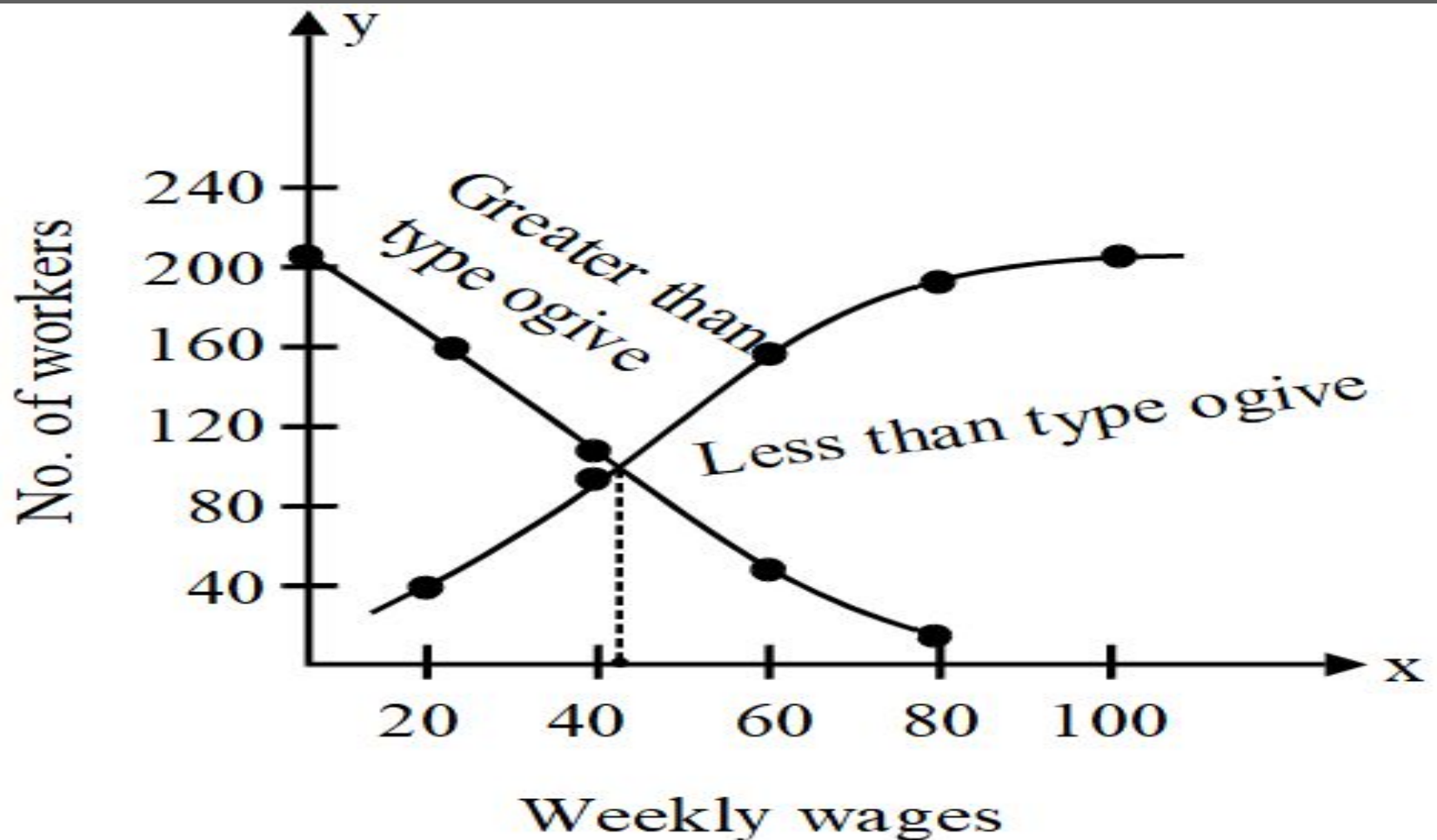


Box Plot

OGIVE (CUMULATIVE FREQUENCY CURVE)

1. *Less than Ogive*: Plot the points with the upper limits of the class as abscissa and the corresponding less than cumulative frequencies as ordinates. The points are joined by free hand smooth curve to give less than cumulative frequency curve or the less than Ogive. It is a rising curve.
2. *Greater than Ogive*: Plot the points with the lower limits of the classes as abscissa and the corresponding Greater than cumulative frequencies as ordinates. Join the points by a free hand smooth curve to get the “More than Ogive”. It is a falling curve.

OGIVE (CUMULATIVE FREQUENCY CURVE)



OGIVE (CUMULATIVE FREQUENCY CURVE)

Construct the combined ogive from the data given in the table

Markes	No. of students
0-10	3
10-20	8
20-30	17
30-40	29
40-50	15
50-60	6
60-70	2

OGIVE (CUMULATIVE FREQUENCY CURVE)

Construct the combined ogive from the data given in the table

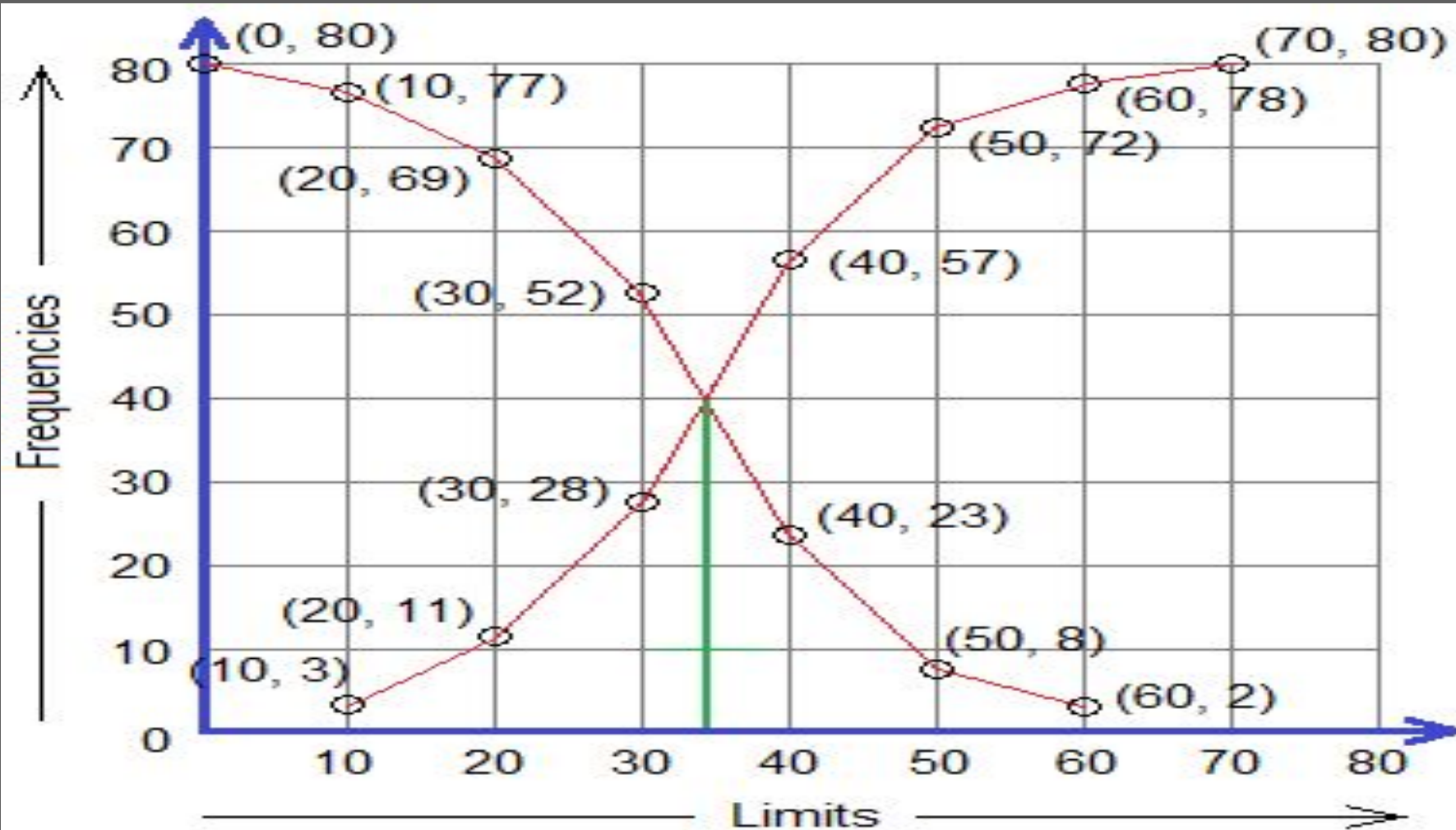
Marks	No. of students
0-10	3
10-20	8
20-30	17
30-40	29
40-50	15
50-60	6
60-70	2

Solution: Here,

Less than and more than cumulative frequency distribution table:

Marks	f	Upper limit	Less than c.f.	Lower limit	More than c.f.
0-10	3	10	3	0	80
10-20	8	20	11	10	77
20-30	17	30	28	20	69
30-40	29	40	57	30	52
40-50	15	50	72	40	23
50-60	6	60	78	50	8
60-70	2	70	80	60	2

Marks	f	Upper limit	Less than c.f.	Lower limit	More than c.f.
0-10	3	10	3	0	80
10-20	8	20	11	10	77
20-30	17	30	28	20	69
30-40	29	40	57	30	52
40-50	15	50	72	40	23
50-60	6	60	78	50	8
60-70	2	70	80	60	2



NUMERICAL PRESENTATION

A fundamental concept in summary statistics is that of a *central value* for a set of observations and the extent to which the central value characterizes the whole set of data. Measures of central value such as the mean or median must be coupled with measures of data dispersion (e.g., average distance from the mean) to indicate how well the central value characterizes the data as a whole.

To understand how well a central value characterizes a set of observations, let us consider the following two sets of data:

A: 30, 50, 70

B: 40, 50, 60

The mean of both two data sets is 50. But, the distance of the observations from the mean in data set A is larger than in the data set B. Thus, the mean of data set B is a better representation of the data set than is the case for set A.

MEASURES OF CENTRAL TENDENCY

- **Measure of Central Tendency:** Usually when two or more different data sets are to be compared it is necessary to condense the data, but for comparison the condensation of data set into a frequency distribution and visual presentation are not enough. It is then necessary to summarize the data set in a single value. Such a value usually somewhere in the center and represent the entire data set and hence it is called measure of central tendency or averages. Since a measure of central tendency (i.e. an average) indicates the location or the general position of the distribution on the X-axis therefore it is also known as a measure of location or position.

METHODS OF CENTER MEASUREMENT

Center measurement is a summary measure of the overall level of a dataset

Commonly used methods are **mean, median, mode, geometric mean** etc.

Mean: Summing up all the observation and dividing by number of observations. Mean of 20, 30, 40 is $(20+30+40)/3 = 30$.

Notation : Let $x_1, x_2, \dots, x_n$ are n observations of a variable x . Then the mean of this variable,

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Arithmetic Mean or Simply Mean: “A value obtained by dividing the sum of all the observations by the number of observation is called arithmetic Mean”

$$\text{Mean} = \frac{\text{Sum of All observation}}{\text{Number of observation}}$$

Methods	Ungrouped data	Grouped data
Direct Method	$\bar{x} = \frac{\sum x_i}{n}$	$\bar{x} = \frac{\sum fx}{n}$; Here $n = \sum f$
Short cut Method	$\bar{x} = A + \frac{\sum D}{n}$	$\bar{x} = A + \frac{\sum fD}{n}$; Here $n = \sum f$
	Where $D = X_i - A$ and A is the provisional or assumed mean.	
Step deviation Method	$\bar{x} = A + \frac{\sum u}{n} \times h$	$\bar{x} = A + \frac{\sum fu}{n} \times h$; Here $n = \sum f$
	Where $u = \frac{X_i - A}{h}$ and h is the common width of the class intervals	

Calculate the arithmetic mean for the following the marks obtained by 9 students are given below:

Using formula of arithmetic mean for ungrouped data:

x_i
45
32
37
46
39
36
41
48
36
$\sum_{i=1}^n x_i = 360$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$n = 9$$

$$\bar{x} = \frac{360}{9} = 40 \text{ marks}$$

✧ Calculate the arithmetic mean for the following data given below:

◆ Using formula of **direct method** of arithmetic mean for grouped data:

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}, \quad n = \sum_{i=1}^n f_i$$

The weight recorded to the nearest grams of 60 apples picked out at random from a consignment are given below:

106	107	76	82	109	107	115	93	187	95	123	125
111	92	86	70	126	68	130	129	139	119	115	128
100	186	84	99	113	204	111	141	136	123	90	115
98	110	78	185	162	178	140	152	173	146	158	194
148	90	107	181	131	75	184	104	110	80	118	82

Weight (grams)	Frequency
65----84	09
85----104	10
105----124	17
125----144	10
145----164	05
165----184	04
185----204	05

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	$f_i x_i$
65----84	$(65 + 84)/2 = 74.5$	09	$9 \times 74.5 = 670.5$
85----104	94.5	10	945.0
105----124	114.5	17	1946.5
125----144	134.5	10	1345.0
145----164	154.5	05	772.5
165----184	174.5	04	698.0
185----204	194.5	05	972.5
		$\sum_{i=1}^n f_i = 60$	$\sum_{i=1}^n f_i x_i = 7350.0$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}, \quad n = \sum_{i=1}^n f_i$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{7350.0}{60} = 122.5 \text{ grams (Answer).}$$

- ◆ Using formula of **short cut method** of **arithmetic mean** for **grouped data**:

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i D_i}{\sum_{i=1}^n f_i}, \quad n = \sum_{i=1}^n f_i$$

Where $D_i = X_i - A$ and A is the provisional or assumed mean

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	$D_i = X_i - A$ $A = 114.5$	$f_i D_i$
65----84	$(65 + 84)/2 = 74.5$	09	-40	-360
85----104	94.5	10	-20	-200
105----124	<u>114.5</u>	<u>17</u>	0	0
125----144	134.5	10	20	200
145----164	154.5	05	40	200
165----184	174.5	04	60	240
185----204	194.5	05	80	400
		$\sum_{i=1}^n f_i = 60$		$\sum_{i=1}^n f_i D_i = 480$

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i D_i}{\sum_{i=1}^n f_i} = 114.5 + \frac{480}{60} = 122.5 \text{ grams (Answer).}$$

- ◆ Using formula of **step deviation method** of **arithmetic mean** for **grouped data**:

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i u_i}{\sum_{i=1}^n f_i} \times h, \quad u_i = \frac{x_i - A}{h}, \text{ where } h \text{ is the width of the class interval:}$$

Weight (grams)	Midpoints (x_i)	Frequency (f_i)	$u_i = \frac{X_i - A}{h}$ $A = 114.5, h=20$	$f_i u_i$
65----84	$(65 + 84)/2 = 74.5$	09	-2	-18
85----104	94.5	10	-1	-10
105----124	<u>114.5</u>	<u>17</u>	0	0
125----144	134.5	10	1	10
145----164	154.5	05	2	10
165----184	174.5	04	3	12
185----204	194.5	05	4	20
		$\sum_{i=1}^n f_i = 60$		$\sum_{i=1}^n f_i u_i = 24$

$$\bar{x} = A + \frac{\sum_{i=1}^n f_i u_i}{\sum_{i=1}^n f_i} \times h = 114.5 + \frac{24}{60} \times 20 = 114.5 + 08 = 122.5 \text{ grams (Answer).}$$

Example 2. Find the mean of the following :

Numbers	8	10	15	20
Frequency	5	8	8	4

Solution. $\sum fx = 8 \times 5 + 10 \times 8 + 15 \times 8 + 20 \times 4 = 40 + 80 + 120 + 80 = 320$

$$\sum f = 5 + 8 + 8 + 4 = 25$$

$$A.M. = \frac{\sum fx}{\sum f} = \frac{320}{25} = 12.8$$

Ans.

Example 3. Find the arithmetic mean for the following distribution:

<i>Class</i>	<i>0 – 10</i>	<i>10 – 20</i>	<i>20 – 30</i>	<i>30 – 40</i>	<i>40 – 50</i>
<i>Frequency</i>	<i>7</i>	<i>8</i>	<i>20</i>	<i>10</i>	<i>5</i>

Solution. Let assumed mean (a) = 25.

<i>Class</i>	<i>Mid-value</i> x	<i>Frequency</i> f	$x - 25 = d$	fd
0 – 10	5	7	-20	-140
10 – 20	15	8	-10	-80
20 – 30	25	20	0	0
30 – 40	35	10	+ 10	+ 100
40 – 50	45	5	+ 20	+ 100
Total		50		- 20

$$A M = a + \frac{\sum fd}{\sum f} = 25 + \frac{-20}{50} = 24.6$$

Ans.

Example 4. Find the arithmetic mean of the data given in example 3 by step deviation method

Solution. Let $a = 25$

Class	Mid-value x	frequency f	$D = \frac{x-a}{i}$	$f \cdot D$
0 – 10	5	7	-2	-14
10 – 20	15	8	-1	-8
20 – 30	25	20	0	0
30 – 40	35	10	+ 1	+ 10
40 – 50	45	5	+ 2	+ 10
Total		50		- 2

$$A.M. = a + \frac{\sum fD}{\sum f} i = 25 + \frac{-2}{50} \times 10 = 24.6$$

Ans.

- 20. Commissions Earned** This frequency distribution represents the commission earned (in dollars) by 100 salespeople employed at several branches of a large chain store. *a. 180.3 b. 177–185*

Class limits	Frequency
150–158	5
159–167	16
168–176	20
177–185	21
186–194	20
195–203	15
204–212	3

- 21. Copier Service Calls** This frequency distribution represents the data obtained from a sample of 75 copying machine service technicians. The values represent the days between service calls for various copying machines. *a. 23.7 b. 21.5–24.5*

Class boundaries	Frequency
15.5–18.5	14
18.5–21.5	12
21.5–24.5	18
24.5–27.5	10
27.5–30.5	15
30.5–33.5	6

- 15. Percentage of Foreign-Born People** The percentage of foreign-born population for each of the 50 states is represented below. Do you think the mean is the best average for this set of data? Explain. *a. 6.5 b. 0.8–4.4. Probably not—data are “top heavy.”*

Percentage	Frequency
0.8–4.4	26
4.5–8.1	11
8.2–11.8	4
11.9–15.5	5
15.6–19.2	2
19.3–22.9	1
23.0–26.6	1

- 14. Automobile Fuel Efficiency** Thirty automobiles were tested for fuel efficiency (in miles per gallon). This frequency distribution was obtained. (The data in this exercise will be used in Exercise 20 in Section 3–2.) *a. 19.7 b. 17.5–22.5*

Class boundaries	Frequency
7.5–12.5	3
12.5–17.5	5
17.5–22.5	15
22.5–27.5	5
27.5–32.5	2

- 17. Percentage of College-Educated Population over 25** Below are the percentages of the population over 25 years of age who have completed 4 years of college or more for the 50 states and the District of Columbia. Find the mean and modal class. *a. 26.7 b. 24.2–28.6*

Percentage	Frequency
15.2–19.6	3
19.7–24.1	15
24.2–28.6	19
28.7–33.1	6
33.2–37.6	7
37.7–42.1	0
42.2–46.6	1

Source: *New York Times Almanac*.

- 13. Hourly Compensation for Production Workers** The hourly compensation costs (in U.S. dollars) for production workers in selected countries are represented below.

Class	Frequency
2.48–7.48	7
7.49–12.49	3
12.50–17.50	1
17.51–22.51	7
22.52–27.52	5
27.53–32.53	5

a. 17.68 b. 2.48–7.48 and 17.51–22.51. Group mean is less.

Compare the mean of these grouped data to the U.S. mean of \$21.97.

METHODS OF CENTER MEASUREMENT

“MEDIAN”

Median: *“when the observation are arranged in ascending or descending order, then a value, that divides a distribution into equal parts, is called median”*

The middle value in an ordered sequence of observations. That is, to find the median we need to order the data set and then find the middle value. In case of an even number of observations the average of the two middle most values is the median. For example, to find the median of {9, 3, 6, 7, 5}, we first sort the data giving {3, 5, 6, 7, 9}, then choose the middle value 6. If the number of observations is even, e.g., {9, 3, 6, 7, 5, 2}, then the median is the average of the two middle values from the sorted sequence, in this case, $(5 + 6) / 2 = 5.5$.

Median in case of Ungrouped Data

In this case we first arrange the observations in **increasing** or **decreasing** order then we use the following formulae for Median:

If “n” is odd

$$\text{Median} = \text{size of } \left(\frac{n+1}{2} \right) \text{th observation}$$

If “n” is even

$$\text{Median} = \frac{\text{size of } \left\{ \left(\frac{n}{2} \right) \text{th} + \left(\frac{n}{2} + 1 \right) \text{th} \right\} \text{ observation}}{2}$$

Calculate the median for the following the marks obtained by 9 students are given below:

x_i	45	32	37	46	39	36	41	48
-------	----	----	----	----	----	----	----	----

Calculate the median for the following the marks obtained by 9 students are given below:

x_i 45 32 37 46 39 36 41 48

Arrange the data in ascending order

32, 36, 37, 39, 41, 45, 46, 48.

$n = 8$ “ n ” is odd

If “ n ” is odd

Median = size of $\left(\frac{n+1}{2}\right)$ th observation

ascending order

32, 36, 36, 37, 39, 41, 45, 46, 48. $n = 9$ “ n ” is odd

Median = Size of $\left(\frac{n+1}{2}\right)^{th}$ observation

Median = Size of $\left(\frac{9+1}{2}\right)^{th}$ observation

Median = Size of 5^{th} observation

Median = 39 (Answer).

Calculate the median for the following the marks obtained by 10 students are given below:

x_i	45	32	37	46	39	36	41	48	36	50
-------	----	----	----	----	----	----	----	----	----	----

Calculate the median for the following the marks obtained by 10 students are given below:

x_i	45	32	37	46	39	36	41	48	36	50
-------	----	----	----	----	----	----	----	----	----	----

Arrange the data in ascending order

32, 36, 36, 37, 39, 41, 45, 46, 48, 50.

$$n = 10$$

“n” is even

Arrange the data in ascending order

32, 36, 36, 37, 39, 41, 45, 46, 48, 50. $n = 10$ “ n ” is even

$$\text{Median} = \frac{\text{Size of } \left\{ \left(\frac{n}{2} \right)^{\text{th}} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \right\} \text{ observation}}{2}$$

$$\text{Median} = \frac{\text{Size of } \left\{ \left(\frac{10}{2} \right)^{\text{th}} + \left(\frac{10}{2} + 1 \right)^{\text{th}} \right\} \text{ observation}}{2}$$

$$\text{Median} = \frac{\text{Size of } \{ 5^{\text{th}} + 6^{\text{th}} \} \text{ observation}}{2}$$

$$\text{Median} = \frac{39 + 41}{2} = 40 \quad (\text{Answer}).$$

Median in case of Discrete Grouped Data

Median in case of Discrete Grouped Data

In case of discrete grouped data, first we find the cumulative frequencies and then use the following formula for Median:

$$\text{Median} = \text{size of } \left(\frac{n+1}{2} \right) \text{th observation}$$

$$\text{Here } n = \sum f$$

Numerical examples: The following distribution relates to the number of assistants in 50 retail establishments.

No.of assistant	0	1	2	3	4	5	6	7	8	9
f	3	4	6	7	10	6	5	5	3	1

Numerical examples: The following distribution relates to the number of assistants in 50 retail establishments.

No.of assistant	0	1	2	3	4	5	6	7	8	9
f	3	4	6	7	10	6	5	5	3	1

No. of assistants	f_i	Cumulative frequency ($c.f$)
0	3	3
1	4	7
2	6	13
3	7	20
4	10	30
5	6	36
6	5	41
7	5	46
8	3	49
9	1	50
	$\sum_{i=1}^n f_i = 50$	

$$n = \sum_{i=1}^n f_i = 50 \quad \text{"n" is even}$$

$$\text{Median} = \frac{\text{Size of } \left\{ \left(\frac{n}{2} \right)^{\text{th}} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \right\} \text{ observation}}{2}$$

$$\text{Median} = \frac{\text{Size of } \left\{ \left(\frac{50}{2} \right)^{\text{th}} + \left(\frac{50}{2} + 1 \right)^{\text{th}} \right\} \text{ observation}}{2}$$

No. of assistants	f_i	Cumulative frequency ($c.f$)
0	3	3
1	4	7
2	6	13
3	7	20
4	10	30
5	6	36
6	5	41
7	5	46
8	3	49
9	1	50
	$\sum_{i=1}^n f_i = 50$	

$$\text{Median} = \frac{\text{Size of } \{25^{\text{th}} + 26^{\text{th}}\} \text{ observation}}{2}$$

$$\text{Median} = \frac{4 + 4}{2} = 4 \quad (\text{Answer}).$$

Median in case of Continues Grouped Data

Median in case of continuous Grouped Data

In continuous grouped data, when we are finding median, we first construct the class boundaries if the classes are discontinuous. Then we find cumulative frequencies and then we use the following two steps:

- First we determine the median class using $n/2$.
- When the median class is determined, then the following formula is used to find the value of median. i.e.

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - C \right); \quad \text{Here } n = \sum f$$

Where l = lower class boundary of the median class

h = width of the median class

f = frequency of the median class

C = cumulative frequency of the class preceding the median class.

Numerical Example: Find the median, for the distribution of examination marks given below:

Marks	30--39	40--49	50--59	60--69	70--79	80--89	90--99
Number of students	08	87	190	304	211	85	20

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - C \right) \quad \text{here } n = \sum_{i=1}^n f_i$$

Class boundaries	Midpoints (x_i)	Frequency (f_i)	Cumulative frequency ($c.f$)
29.5---39.5	34.5	8	8
39.5---49.5	44.5	87	95
49.5---59.5	54.5	190	285
59.5---69.5	64.5	304	589
69.5---79.5	74.5	211	800
79.5---89.5	84.5	85	885
89.5---99.5	94.5	20	905
		$\sum_{i=1}^n f_i = 905$	

$$\frac{n}{2} = \frac{905}{2} = 452.5^{\text{th}} \text{ student which corresponds to marks in the class } 59.5\text{---}69.5$$

Therefore

$$\text{Median} = l + \frac{h}{f} \left(\frac{n}{2} - C \right) = 59.5 + \frac{10}{304} \left(\frac{905}{2} - 285 \right) = 59.5 + \frac{10}{304} (452.5 - 285)$$

$$\text{Median} = 59.5 + \frac{1675}{304}, \text{ Median} = 59.5 + 5.5, \text{ Median} = 65 \text{ marks} \quad (\text{Answer}).$$

Example 6. Find the value of Median from the following data:

No. of days for which absent (less than)	5	10	15	20	25	30	35	40	45
No. of students	29	224	465	582	634	644	650	653	655

Example 6. Find the value of Median from the following data:

No. of days for which absent (less than)	5	10	15	20	25	30	35	40	45
No. of students	29	224	465	582	634	644	650	653	655

Solution. The given cumulative frequency distribution will first be converted into ordinary frequency as under

<i>Class- Interval</i>	<i>Cumulative frequency</i>	<i>Ordinary frequency</i>
0 – 5	29	$29 = 29$
5 – 10	224	$224 - 29 = 195$
10 – 15	465	$465 - 224 = 241$
15 – 20	582	$582 - 465 = 117$
20 – 25	634	$634 - 582 = 52$
25 – 30	644	$644 - 634 = 10$
30 – 35	650	$650 - 644 = 6$
35 – 40	653	$653 - 650 = 3$
40 – 45	655	$655 - 653 = 2$

$$\text{Median} = \text{size of } \frac{655}{2} \text{ or } 327.5\text{th item}$$

327.5th item lies in 10-15 which is the median class.

$$M = l + \frac{\frac{N}{2} - C}{f} i$$

where l stands for lower limit of median class,

N stands for the total frequency,

C stands for the cumulative frequency just preceeding the median class,

i stands for class interval

f stands for frequency for the median class.

$$\begin{aligned}\text{Median} &= 10 + \frac{\frac{655}{2} - 224}{241} \times 5 \\ &= 10 + \frac{103.5 \times 5}{241} = 10 + 2.15 = 12.15\end{aligned}$$

Ans.

MODE

Mode is defined to be the size of the variable which occurs most frequently.

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Mode is defined to be the size of the variable which occurs most frequently.

Example: Find the mode of the following items : 0, 1, 6, 7, 2, 3, 7, 6, 6, 2, 6, 0, 5, 6, 0.

Solution.

6 occurs 5 times and no other item occurs 5 or more than 5 times, hence the mode is 6.

MODE

For grouped data

$$Mode = l + \frac{f - f_{-1}}{2f - f_{-1} - f_1} \cdot i$$

Find the mode from the following data:

Age	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30	30 – 36	36 – 42
Frequency	6	11	25	35	18	12	6

Find the mode from the following data:

Age	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30	30 – 36	36 – 42
Frequency	6	11	25	35	18	12	6

<i>Age</i>	<i>Frequency</i>	<i>Cumulative frequency</i>
0 – 6	6	6
6 – 12	11	17
12 – 18	$25 = f_{-1}$	42
18 – 24	$35 = f$	77
24 – 30	18 $= f_1$	95
30 – 36	12	107
36 – 42	6	113

Find the mode from the following data:

Age	0 – 6	6 – 12	12 – 18	18 – 24	24 – 30	30 – 36	36 – 42
Frequency	6	11	25	35	18	12	6

<i>Age</i>	<i>Frequency</i>	<i>Cumulative frequency</i>
0 – 6	6	6
6 – 12	11	17
12 – 18	$25 = f_{-1}$	42
18 – 24	$35 = f$	77
24 – 30	$18 = f_1$	95
30 – 36	12	107
36 – 42	6	113

$$Mode = l + \frac{f - f_{-1}}{2f - f_{-1} - f_1} \cdot i$$

$$\text{Mode} = l + \frac{f - f_{-1}}{2f - f_{-1} - f_1} \times i$$

$$= 18 + \frac{35 - 25}{70 - 25 - 18} \times 6$$

$$= 18 + \frac{60}{27} = 18 + 2.22 = 20.22$$

Ans.

MEAN DEVIATION

It is the mean of the absolute values of the deviations of a given set of numbers from their arithmetic mean.

If $x_1, x_2, x_3, \dots, x_n$ be a set of numbers with frequencies $f_1, f_2, \dots, f_n$ respectively. Let $\bar{x}$ be the arithmetic mean of the numbers $x_1, x_2, \dots, x_n$, then

$$\text{Mean deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

Find the mean deviation of the following frequency distribution

Class	0 – 6	6 –12	12 –18	18 – 24	24 –30
Frequency	8	10	12	9	5

Find the mean deviation of the following frequency distribution

Class	0 – 6	6 –12	12 –18	18 – 24	24 –30
Frequency	8	10	12	9	5

$$\text{Mean deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$\text{Mean} = a + \frac{\sum fd}{\sum f}$$

Solution. Let $a = 15$

Class	Mid- value x	Frequency f	$d = x - a$	fd	$ x - 14 $	$f x - 14 $
0–6	3	8	–12	–96	11	88
6–12	9	10	–6	–60	5	50
12–18	15	12	0	0	1	12
18–24	21	9	+ 6	54	7	63
24–30	27	5	+ 12	60	13	65
Total		44		– 42		278

$$\text{Mean} = a + \frac{\sum fd}{\sum f} = 15 - \frac{42}{44} = 14 \text{ nearly}$$

$$\text{Average deviation} = \frac{\sum f|x - \bar{x}|}{\sum f} = \frac{278}{44} = 6.3$$

Ans.